Data Visualization

Computing Fundamentals for Social Sciences

Claudia Pigini

Outline

What is a Frequency Distribution?

- Shows how often each value or category occurs.
- Helps understand the structure of your data.
- Provides a first step before plots and statistical analysis.

Types of Variables

- **Qualitative (Categorical)**
  - Nominal: no inherent order (e.g., Gender)
  - Ordinal: ordered categories (e.g., Education Level)
- **Quantitative (Numerical)**
  - Discrete: countable numbers (e.g., Number of children)
  - Continuous: measurable quantities (e.g., Height, Weight)

Frequency Table - Categorical Variable

```
data <- read.csv("data.csv")

# Frequency table
table(data$Gender)
```

- Count of each category.
- Provides a simple overview of categorical distributions.

Relative Frequency and Percentages

```r
# Relative frequency
prop.table(table(data$Gender))

# Percentages
round(prop.table(table(data$Gender)) * 100, 1)
```

Bar Plot - Categorical Variable

```
library(ggplot2)

ggplot(data, aes(x = Gender)) +
  geom_bar(fill = "skyblue") +
  labs(title = "Gender Distribution",
       x = "Gender",
       y = "Count")
```

Frequency Table - Discrete Variable

```
table(data$Children)
```

Bar Plot - Discrete Variable

```
ggplot(data, aes(x = factor(Children))) +
  geom_bar(fill = "orange") +
  labs(title = "Number of Children Distribution",
       x = "Children",
       y = "Count")
```

Grouping Continuous Variables

```
breaks <- seq(min(data$Height), max(data$Height), by = 5)
data$HeightGroup <- cut(data$Height, breaks = breaks,
    include.lowest = TRUE)
table(data$HeightGroup)
```

- Grouping into intervals helps summarize continuous data.
- Frequency tables for intervals make trends visible.

Histogram - Continuous Variable

```r
ggplot(data, aes(x = Height)) +
  geom_histogram(binwidth = 5, fill = "green", color = "
      black") +
  labs(title = "Height Histogram",
       x = "Height (cm)",
       y = "Count")
```

Relative Frequency Histogram

```r
ggplot(data, aes(x = Height, y = ..density..)) +
  geom_histogram(binwidth = 5, fill = "lightgreen", color =
     "black") +
  labs(title = "Height Relative Frequency Histogram",
       x = "Height (cm)",
       y = "Density")
```

Violin Plot - Continuous by Group

```
ggplot(data, aes(x = Group, y = Score, fill = Group)) +
  geom_violin(trim = FALSE, color = "black") +
  geom_boxplot(width = 0.1, fill = "white") +
  labs(title = "Score Distribution by Group",
       x = "Group",
       y = "Score")
```

Scatterplot - Relationship Between Variables

```
ggplot(data, aes(x = VariableX, y = VariableY)) +
  geom_point(color = "darkgreen", size = 3) +
  labs(title = "Scatterplot Example",
       x = "Variable X",
       y = "Variable Y")
```

- Identify correlations or patterns.
- Check for outliers or clusters.

Key Points

- Frequency tables summarize data: counts, percentages.
- Bar plots: categorical and discrete data.
- Histograms: continuous data.
- Violin plots: distributions by group.
- Scatterplots: relationships between two variables.

Practice Exercises

1. Load the CSV dataset and examine it.
2. Create frequency tables for categorical and discrete variables.
3. Compute percentages for categories.
4. Create bar plots, histograms, and violin plots.
5. Make a scatterplot and interpret the relationship.